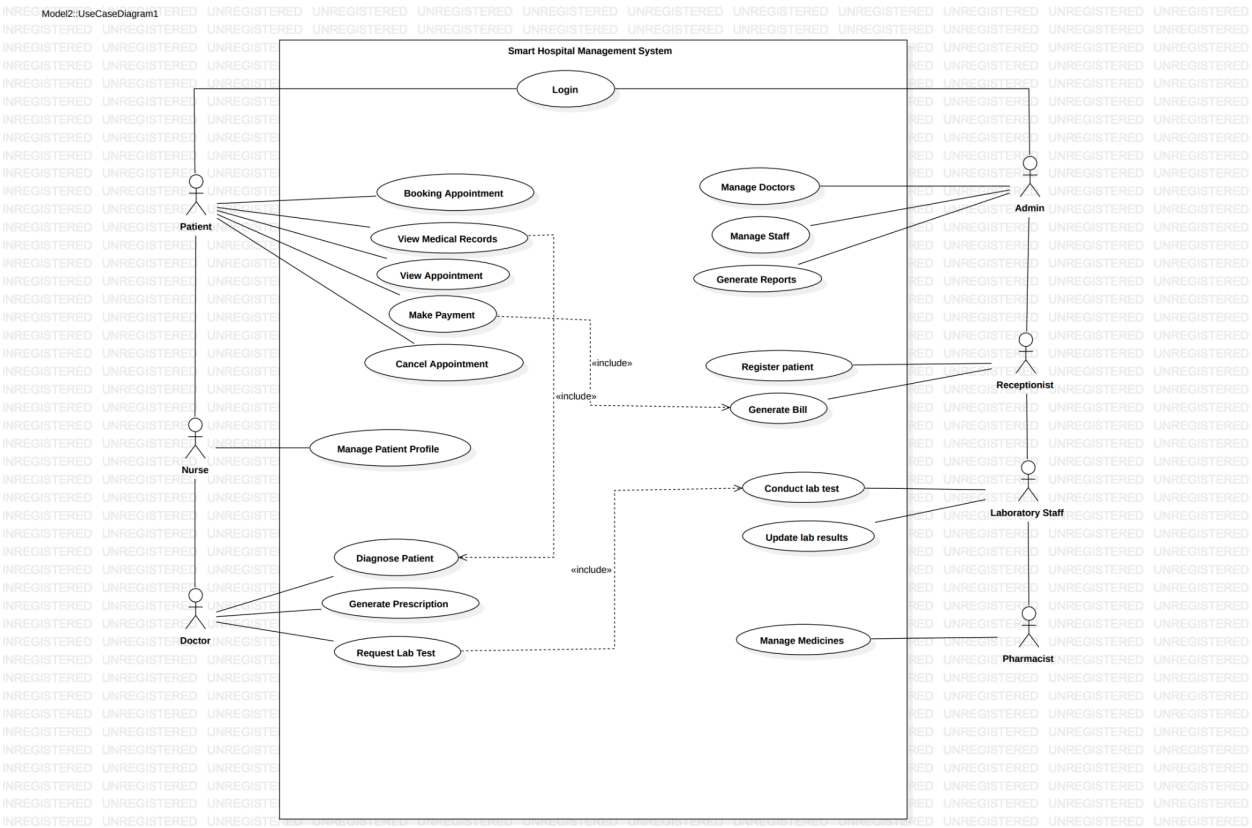


Actor Descriptions

- **Patient:** Books, views, and cancels appointments, accesses medical records, and makes payments.
- **Doctor:** Diagnoses patients, generates prescriptions, and requests laboratory tests.
- **Nurse:** Manages and updates patient profiles and assists in patient care.
- **Admin:** Manages doctors and staff and generates hospital reports.
- **Receptionist:** Registers patients and generates bills.
- **Laboratory Staff:** Conducts laboratory tests and updates test results.
- **Pharmacist:** Manages medicines and pharmacy-related information.

Use Case Descriptions

- **Login:** Allows all authorized users to securely access the system according to their roles.
- **Booking Appointment:** Allows patients to schedule an appointment with a doctor.
- **View Medical Records:** Allows patients to view their stored medical information and records.
- **View Appointment:** Allows patients to view their scheduled appointment details.
- **Make Payment:** Allows patients to pay their hospital bills.
- **Cancel Appointment:** Allows patients to cancel a previously booked appointment.
- **Manage Patient Profile:** Allows nurses to view and update patient profile information.
- **Diagnose Patient:** Allows doctors to examine patient information and record a diagnosis.
- **Generate Prescription:** Allows doctors to create prescriptions for diagnosed patients.
- **Request Lab Test:** Allows doctors to request laboratory tests for patients.
- **Manage Doctors:** Allows administrators to add, update, or manage doctor information.
- **Manage Staff:** Allows administrators to manage hospital staff information.
- **Generate Reports:** Allows administrators to generate hospital-related reports.
- **Register Patient:** Allows receptionists to register new patients in the system.
- **Generate Bill:** Allows receptionists to generate bills for patients.
- **Conduct Lab Test:** Allows laboratory staff to perform requested laboratory tests.
- **Update Lab Results:** Allows laboratory staff to enter and update test results.
- **Manage Medicines:** Allows pharmacists to manage available medicines and medicine information.

Connections / Relationships

- **Login → All Actors:** All actors must log in to access the system according to their permissions.
- **Patient → Booking Appointment:** The patient initiates the appointment booking process.
- **Patient → View Medical Records:** The patient accesses their medical records.
- **Patient → View Appointment:** The patient checks their appointment details.
- **Patient → Make Payment:** The patient makes payment for hospital services.
- **Patient → Cancel Appointment:** The patient cancels a booked appointment.
- **Nurse → Manage Patient Profile:** The nurse manages patient profile information.
- **Doctor → Diagnose Patient:** The doctor records the patient's diagnosis.
- **Doctor → Generate Prescription:** The doctor generates a prescription after diagnosis.
- **Doctor → Request Lab Test:** The doctor requests laboratory tests for the patient.
- **Admin → Manage Doctors:** The admin manages doctor information.
- **Admin → Manage Staff:** The admin manages hospital staff information.
- **Admin → Generate Reports:** The admin generates hospital reports.
- **Receptionist → Register Patient:** The receptionist registers new patients.
- **Receptionist → Generate Bill:** The receptionist generates patient bills.

- **Laboratory Staff → Conduct Lab Test:** Laboratory staff perform requested tests.
- **Laboratory Staff → Update Lab Results:** Laboratory staff record the test results.
- **Pharmacist → Manage Medicines:** The pharmacist manages medicine information.
- **View Medical Records → Diagnose Patient (`<<include>>`):** Patient medical records are included when diagnosing a patient.
- **Make Payment → Generate Bill (`<<include>>`):** Bill generation is included as part of the payment process.
- **Request Lab Test → Conduct Lab Test (`<<include>>`):** A requested laboratory test leads to the test being conducted.